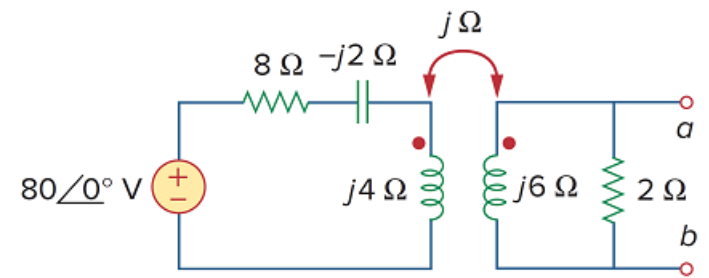


Problem

Obtain the Norton equivalent at terminals $a-b$ of the circuit in Fig. 13.85.

Figure 13.85



Step-by-step solution

Step 1 of 14

Consider the following circuit:

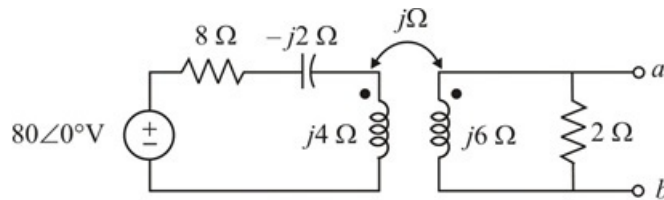


Figure 1

Step 2 of 14

Short circuit the terminals a and b . Determine the Norton current I_N .

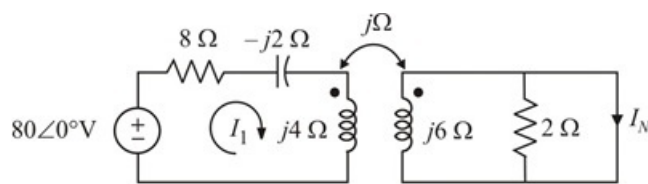


Figure 2

Step 3 of 14

Since, the 2Ω resistor is in parallel with a short circuit.

Redraw the circuit as shown in figure 3.

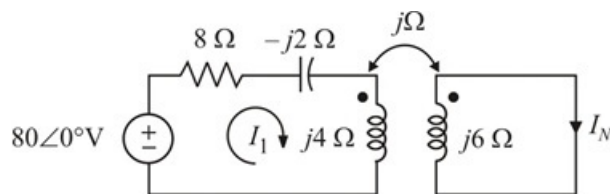


Figure 3

Step 4 of 14

Apply Kirchhoff's voltage law to the primary loop.

$$-80\angle 0^\circ + (8 - j2 + j4)I_1 - (j1)I_N = 0$$

$$(8 + j2)I_1 - (j1)I_N = 80 \dots\dots (1)$$

Writing KVL equation across the secondary loop,

$$(j6)I_N - (j1)I_1 = 0$$

$$I_1 = 6I_N \dots\dots (2)$$

Step 5 of 14

Consider equation (2).

Substitute $6I_N$ for I_1 in equation (1).

$$(8 + j2)(6I_N) - (j1)I_N = 80$$

$$(48 + j12 - j1)I_N = 80$$

$$I_N = \frac{80}{48 + j11}$$

$$I_N = \frac{80}{49.244\angle 12.91^\circ}$$

Therefore, the Norton current is $1.6246\angle -12.91^\circ \text{ A}$.

Step 6 of 14

Determine the Norton impedance Z_N .

Short circuit the voltage source on the primary side and place a 1 A test current source across the terminals a and b.

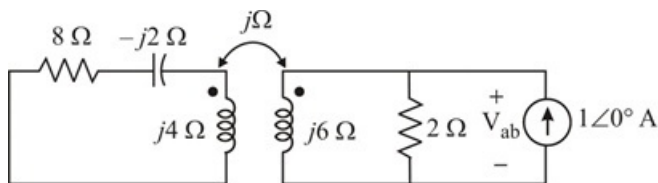


Figure 4

Step 7 of 14

Apply Source transformation technique. Convert the current source to a voltage source with series resistance as shown in figure 5.

$$V = IR$$

$$V = (1\angle 0^\circ)(2)$$

$$V = 2\angle 0^\circ \text{ V}$$

Step 8 of 14

Therefore the circuit is,

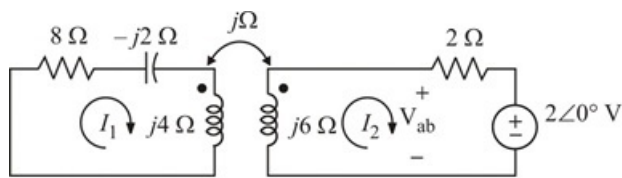


Figure 5

Step 9 of 14

Apply Kirchhoff's voltage law to the primary loop of the circuit shown in figure 5.

$$(8 - j2 + j4)I_1 - (j1)I_2 = 0$$

$$(8 + j2)I_1 - (j1)I_2 = 0$$

$$I_1 = \left(\frac{j1}{8 + j2} \right) I_2 \dots\dots (3)$$

Step 10 of 14

Apply Kirchhoff's voltage law to the secondary loop of the circuit shown in figure 5.

$$(2 + j6)I_2 - (j1)I_1 + 2\angle 0^\circ = 0$$

$$(-j1)I_1 + (2 + j6)I_2 = -2 \dots\dots (4)$$

Step 11 of 14

Recall equation (3).

$$I_1 = \left(\frac{j1}{8 + j2} \right) I_2$$

Substitute $\left(\frac{j1}{8 + j2} \right) I_2$ for I_1 in equation (4).

$$(-j1) \left(\frac{j1}{8 + j2} \right) I_2 + (2 + j6)I_2 = -2$$

$$\frac{1}{(8 + j2)} I_2 + (2 + j6)I_2 = -2$$

$$I_2 [1 + (8 + j2)(2 + j6)] = -2(8 + j2)$$

$$I_2 [1 + 16 + j48 + j4 - 12] = -16 - j4$$

Simplify further,

$$I_2 = \frac{-16 - j4}{5 + j52}$$

$$I_2 = \frac{16.49\angle -165.96^\circ}{52.24\angle 84.51^\circ}$$

$$I_2 = 0.3157\angle -250.47^\circ$$

$$I_2 = 0.3157\angle 109.53^\circ \text{ A}$$

Step 12 of 14

Calculate voltage across the terminals a and b .

$$V_{ab} = 2I_2 + 2\angle 0^\circ$$

Substitute $0.3157\angle 109.53^\circ$ for I_2 .

$$V_{ab} = 2(0.3157\angle 109.53^\circ) + 2$$

$$V_{ab} = 0.6314\angle 109.53^\circ + 2$$

$$V_{ab} = -0.2111 + j0.5951 + 2$$

$$V_{ab} = 1.7889 + j0.5951$$

Simplify further,

$$V_{ab} = 1.885\angle 18.4^\circ \text{ V}$$

Step 13 of 14

Determine the Norton impedance Z_N .

$$Z_N = \frac{V_{ab}}{1\angle 0^\circ}$$

Substitute $1.885\angle 18.4^\circ$ for V_{ab} .

$$Z_N = \frac{1.885\angle 18.4^\circ}{1\angle 0^\circ}$$

Therefore, the Norton impedance is $1.885\angle 18.4^\circ \Omega$.

Step 14 of 14

Draw the Norton equivalent circuit as shown in figure 6.

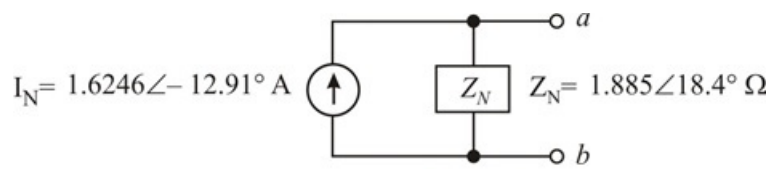


Figure 6